

AI for Model Validation

Locate weakness, investigate repair and establish the evidence.

01 Define and test

Reference evidence, Inner and Outer Arrays, evaluation and Pareto comparison.

02 Search and refine

Ordinary optimization, then structural discovery when the test space is insufficient.

03 Build and verify evidence

Knowledge graphs, claim verification and adversarial testing of agentic systems.

04 Confirm and report

Reserved evaluation, materiality, supported conclusions and report generation.

1. Introduction

Finding model weaknesses and establishing
which can be repaired

From an initial assessment to a focused discovery loop

Aggregate performance does not locate weakness

PREDICTIVE MODEL

An overall score can conceal high prediction loss in a customer group.

Locate the group, measure the excess loss and assess how much of the population is affected.

AGENTIC SYSTEM

An assistant can answer routine questions correctly yet give unsupported answers under misleading prompts or missing evidence.

Identify the task and conditions in which it fails.

Which conditions expose weakness, how severe is it and does it persist?

Locate weakness and investigate repair

1 FIND REGIONS OF WEAKNESS

CLUSTERING OPTIMIZATION

Keep the predictor fixed. Refine customer regions to locate elevated prediction loss.

Measure severity, coverage and support.
Confirm that the weakness persists.

2 FIND WEAKNESS WE CAN REPAIR

MIXTURE OF EXPERTS

Route cases to specialized predictors and investigate whether their predictions improve.

Compare with the baseline in the targeted regions and across the population.

Confirm elevated regional error; assess repair through measured improvement.

Establish what counts as correct

Predictive reference

Historical test observations pair model inputs with known outcomes. A stated loss measures disagreement with the predictions.

Agentic reference

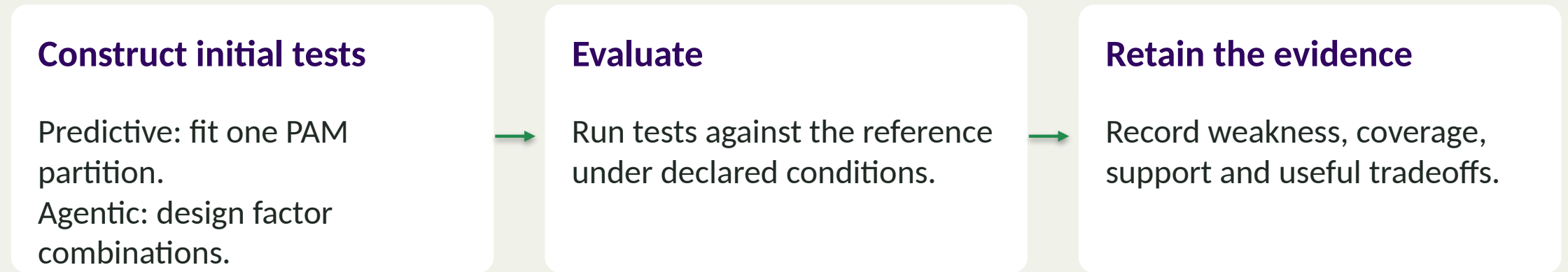
Checked source records or labeled tasks establish the expected answer or permitted action, together with its scope.

Specify the scoring rule and response policy. If required evidence is unavailable, an appropriate answer may be an explicit abstention.

A source must be checked before it can serve as the answer key.

Initial testing can stand alone

Use a planned design before deciding whether further search is needed.

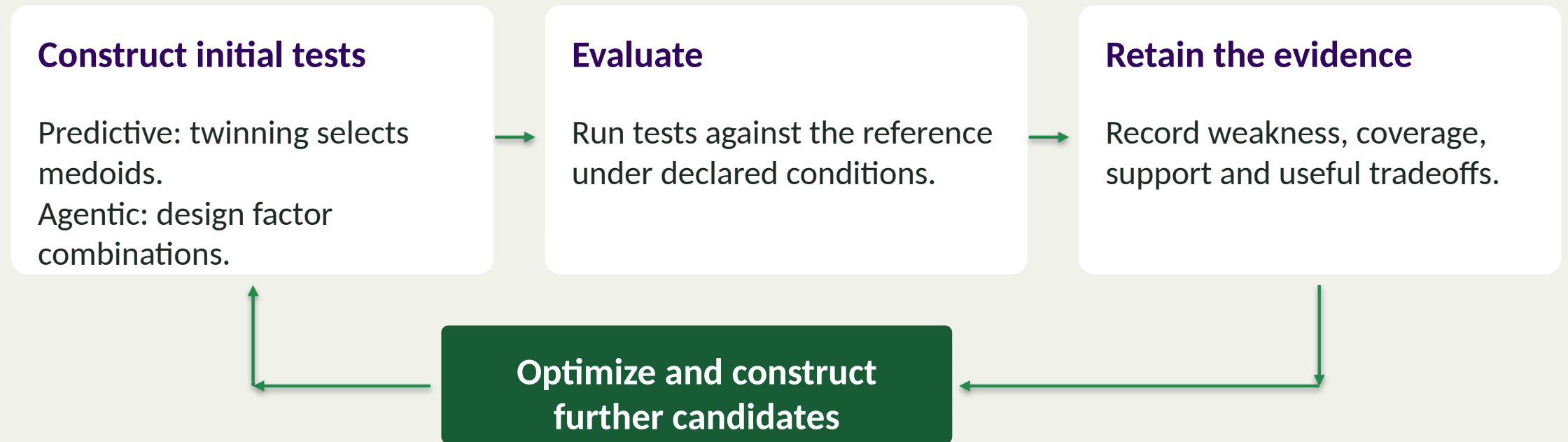


Clustering and design of experiments (DoE) provide an initial assessment.
It can already reveal weaknesses and identify gaps in coverage.

One-shot describes the assessment design; it does not remove confirmation requirements.

Use the results to sharpen the investigation

Evaluate initial candidates, then use measurements to choose further tests.



Sharpen location, severity and extent while retaining enough evidence to assess each test.

When the current test space is insufficient

Search can refine only what the current factors and construction rules allow.
A new failure hypothesis may require changing those rules.

Identify the limit

Example: a one-cell rule excludes deliberately combining two cells.

Propose and check

An LLM or expert proposes a factor or rule. Check constraints and executability.

Resume search

Construct candidates within the admitted structure and measure their outcomes.

Invoke structural reflection when warranted; routine optimization works within the admitted space.

Confirm the weakness and assess the repair

DOES THE WEAKNESS PERSIST?

Freeze the selected region or test rule.
On reserved evidence, measure the weakness, affected share and uncertainty.
State the conditions the finding covers.

DOES THE PROPOSED REPAIR HELP?

Freeze the selected comparator and assessment plan.
Compare with the baseline on reserved cases, both inside and outside the target regions.
Report gains and adverse tradeoffs.

Report two conclusions separately: where the model is weak and what the proposed repair improves.